

Leveraging Transformer Models for Extractive Question Answering in Criminal Report Analysis

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Introduction

Introduction: The Problem



- **Context:** Ecuadorian Prosecutor's Office handled 67,705 robbery incidents in 2024.
- **Need:** Efficient extraction of information from criminal reports for:
 - More effective investigations.
 - Supporting victims' insurance claims.
 - Providing data for public safety planning.
- **Goal:** Adapt Transformer-based models for Extractive Question Answering (EQA) in Ecuadorian legal documents about robbery.

-  **Domain-Adapted BERT model:** First application of transformer-based models for EQA on Spanish-language robbery case reports from Ecuador.
-  **Novel Dataset:** using Generative Artificial Intelligence (GAI) to curate a Spanish-language Stanford Question Answering Dataset (SQuAD) dataset derived from Ecuadorian legal documents.
-  **Potential Impact:** Enhance information retrieval within judicial systems and optimize workloads, improving service delivery to communities.

Related Works

Large Language Models (LLMs) & Generative AI (GAI)

Foundational Concepts

- **Large Language Models (LLMs):** A subset of Foundational Artificial Intelligence (AI) and GAI models, primarily text-based.
 - Leverage transformer architectures [1], [2].
 - Undergo extensive pre-training on textual corpora.
 - Exhibit strong natural language understanding and problem-solving capabilities.
- **GAI:** aims to learn a model distribution $p_{model}(x)$ that approximates the true data distribution $p_{data}(x)$, where x represents observations from the data domain [2].

$$p_{model}(x) \approx p_{data}(x) \quad (1)$$

Question Answering (QA) in Legal Contexts

Complexity and Approaches

- **Legal Question Answering (LQA):** Complex task requiring analysis and nuanced interpretation of extensive legal corpora [3].
- **Traditional Question Answering (QA) Methods:**
 - Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN).
 - Information retrieval approaches like Term Frequency-Inverse Document Frequency (TF-IDF).
- **Modern QA Paradigms:**
 - **EQA:** Identifies and retrieves a direct text span from context.
 - **Generative Question Answering (GQA):** Synthesizes responses using encoder-decoder architectures.

EQA vs. GQA: Key Distinctions

Table: Comparison of EQA and GQA Approaches

Feature	EQA (Extractive)	GQA (Generative)
Output	Direct text span from context	Synthesized, novel response
Architecture	Predicts start/end positions	Encoder-decoder
Context Length	More effective with limited context	Generally outperforms with longer contexts
Fluency	Less fluent (direct extraction)	More fluent and comprehensive
Generalization	Better out-of-domain generalization	Can be limited by training data

Source: Adapted from Luo et al. [4]

Related Works: AI in Legal Domains

State-of-the-Art Applications

- Significant interest in applying AI to legal fields, with transformers leading in Natural Language Processing (NLP) tasks.
- **RoBERTa for Legal QA:** [5] showed RoBERTa outperforming Support Vector Machine (SVM) for legal questions (31% F1-score improvement).
- **Transformer-based Answer Selection:** [6] demonstrated effectiveness in general QA by extracting global information.
- **Legal Information Retrieval:** [7] utilized sentence-transformers for case law retrieval and fine-tuned DeBERTa for legal entailment.

Related Works: AI in Legal Domains (Cont.)

Predicting Judgments & Text Generation

- **Predicting Legal Judgments:** [8] evaluated six transformer models (BERT, XLNet, RoBERTa, DeBERTa, ELECTRA, BigBird) for legal judgment prediction, achieving up to 80% accuracy.
- **Generating Legal Text:** [9] explored transformers for generating judicial opinions and distinguishing human vs. machine-generated legal texts, beneficial for drafting.

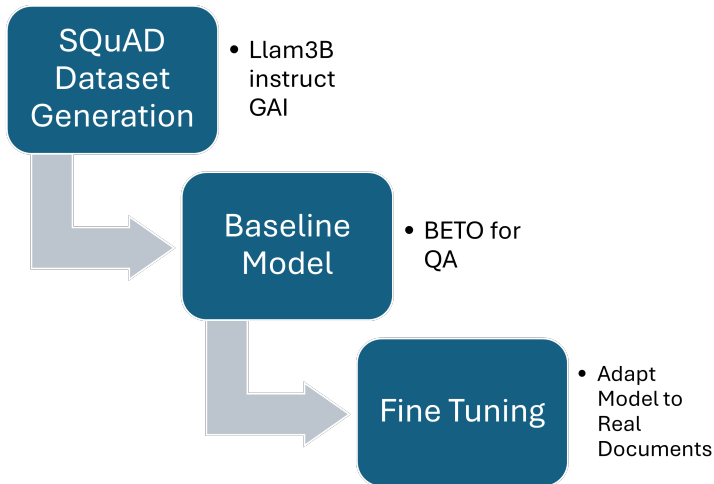
Novelty of the Current Study

- The literature review identifies a critical research gap.
- Our study introduces a novel application of transformer-based models that:
 - Simultaneously addresses **Spanish language** texts.
 - Focuses specifically on **legal documents** (criminal reports).
- This combination fills a unique niche, contributing new insights to Legal NLP (LNLP).

- LLMs and GAI are foundational for advanced NLP, including legal applications.
- Both EQA and GQA have distinct strengths, with EQA being suitable for precise information extraction from defined contexts.
- Transformer models have shown superior performance across various legal NLP tasks, from QA to judgment prediction and text generation.
- Our research uniquely combines Spanish language processing with legal domain adaptation using transformers, addressing a previously unexplored area.

Methodology

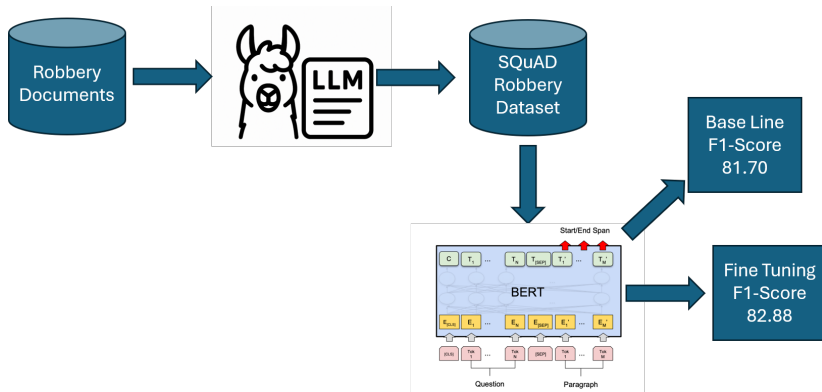
Methodology Overview



Our methodology consists of three phases:

- 1 **Dataset Generation:** Creation of a domain-specific SQuAD-format dataset using GAI.
- 2 **Baseline Model Performance:** Evaluation of a pre-trained BERT model on this dataset.
- 3 **Model Fine-Tuning:** Adaptation of the BERT model to robbery reports.

Phase 1: Robbery Crime QA Dataset Generation



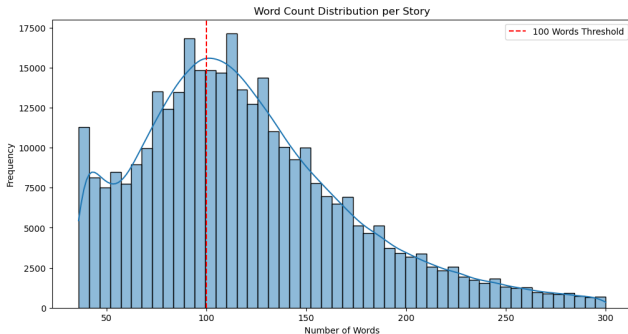
Phase 1: Robbery Crime QA Dataset Generation I

Data Collection and Preprocessing

- Constructed dataset from unlabeled Spanish robbery crime reports.
- Preprocessing: Removed non-alphanumeric characters, converted to lowercase.
- Filtered reports x_i with word counts $82 \leq \Gamma(x_i) \leq 150$ to remove outliers.
- Transformed into SQuAD-compatible format using GAI.

Phase 1: Robbery Crime QA Dataset Generation II

Data Collection and Preprocessing



Distribution of word counts (Histogram)

Phase 1: Robbery Crime QA Dataset Generation

Generative AI Prompt Engineering

- Used **Meta-Llama-3-8B-Instruct** for dataset creation.
- LLMs acted as an expert data science assistant to construct a Spanish SQuAD dataset.
- Prompt received context (crime report) and question (from predefined list).
- Output in json format: context, question, answer_text, answer_start, answer_end, impossible_find_answer.

Phase 1: Robbery Crime QA Dataset Generation

Proposed Questions for the Dataset

Table: Proposed Questions for the Dataset

Question
What objects were stolen?
On what date did the incident occur?
At what time did the robbery happen?
At what address or between which streets did the robbery/incident occur?
What was the dollar value of the stolen or taken objects?

- Initial 1,544 samples generated 7,705 question-answer pairs.
- Identified $\approx 3,000$ invalid cases using `impossible_find_answer` flag.
- Final curated dataset: 4,572 samples (answer length ≤ 17 words).

Phase 1: Robbery Crime QA Dataset Generation

Dataset Train and Test Split

- Random split: 80% for training, 20% for testing.
- Ensures evaluation of model's generalization capability.

Table: Dataset Train and Test Split

Partition	# Samples
Train	3,657
Test	915
Total	4,572

Phase 2: Baseline Model Architecture and Performance

- **Model Selected:**

mrm8488/bert-base-spanish-wwm-cased-finetuned-spa-squad2-es

- Distilled version of BETO (Spanish BERT), fine-tuned on SQuAD-es-v2.0.

- **Architecture (BERT-base configuration):**

- 12 Transformer blocks
- 768-dimensional hidden states
- 12 self-attention heads
- 384 tokens of sequence length

- **Baseline Performance (without additional fine-tuning):** F1-score of 81.70.

Phase 3: Model Fine-Tuning

- All layers are trained.
- Recalculation of start/end indices.
- Truncation applied to context (128-token window) for long texts.
- Small learning rate recommended for fine-tuning to adapt pre-trained knowledge.

Table: Training Parameters

Parameter	Value
Strategy	epoch
Learning rate	2×10^{-5}
Batch Size	64
Epochs	10
Weight Decay	0.01
FP16	True
Metric for best model	F1 (eval)

Experimental Results

Results: Model Performance

Pre-trained Model Performance (Baseline)

Table: Pre-trained Model Performance

Metric	Value
eval_loss	0.9102
eval_exact	53.5519
eval_f1	81.7041
eval_total	915.0000
eval_HasAns_exact	53.5519
eval_HasAns_f1	81.7041
eval_HasAns_total	915.0000

Results: Model Performance

Fine-Tuned Model Performance

Table: Fine-Tuned Model Performance

Metric	Value
eval_loss	1.0088
eval_exact	55.7377
eval_f1	82.8805
eval_total	915.0000
eval_HasAns_exact	55.7377
eval_HasAns_f1	82.8805
eval_HasAns_total	915.0000
epoch	10.0000

- **Improvement:** Fine-tuned model achieved an F1 score of **82.88**, outperforming the pre-trained baseline of 81.70.
- Demonstrates effectiveness of domain-specific fine-tuning.

Conclusions and Future Work

- Demonstrated the effectiveness of transformer-based models for EQA in criminal report analysis.
- Fine-tuned model showed notable improvement (F1 score of 82.88) over the pre-trained model (81.70).
- Model can facilitate extraction of crucial information, supporting legal practitioners.
- Proposed Methodology can be used to fine tune models for LQA in other languages.

Limitations and Future Work

- **Limitations:** Occasional inability to retrieve accurate answers, need for manual verification of results.
- **Future Work:**
 - Explore additional advanced techniques.
 - Incorporate manual labeling by expert peers to improve dataset quality.
 - Utilize additional computational resources for larger datasets.
 - Investigate other NLP techniques in Ecuadorian legal context (e.g., Legal Named Entity Recognition (NER), Legal Document Summarization).
 - Study the proposed methodology in other languages.
- **Overall Goal:** Harness the full potential of AI in supporting the justice system for more efficient case management and resource allocation.

- Faculty of Systems Engineering at Escuela Politécnica Nacional, Quito, Ecuador (academic guidance and financial support).
- Fiscalía General del Estado (cooperation in research development).

Declaration of Competing Interests: The authors have no competing interests to declare.

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